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May 5-6, 2026 Station F, Paris

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GLOBAL OPEN-SOURCE INNOVATION MEETUP

GOSIM Paris 2026

Topic

Introduction of LF AI & Data Foundation and Omni-AI Community

Zhipeng Huang

Chair, LF AI & Data Foundation
Huawei

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Introduction to Omni-AI Community and LFAI & Data Foundation

Zhipeng Huang

- Board chairperson of LFAI&Data Foundation

TAC Chair of Open Model Initiative

Executive member of CCF Open Source Development Committee

Co-lead of the Kubernetes Policy WG, project lead of CNCF Security SIG, founder of the OpenStack Cyborg project, and co-founder of OpenStack Public Cloud WG.

Keynote Speech at various flagship open source summit such as OpenStack Summit, KubeCon and RISC-V Workshop

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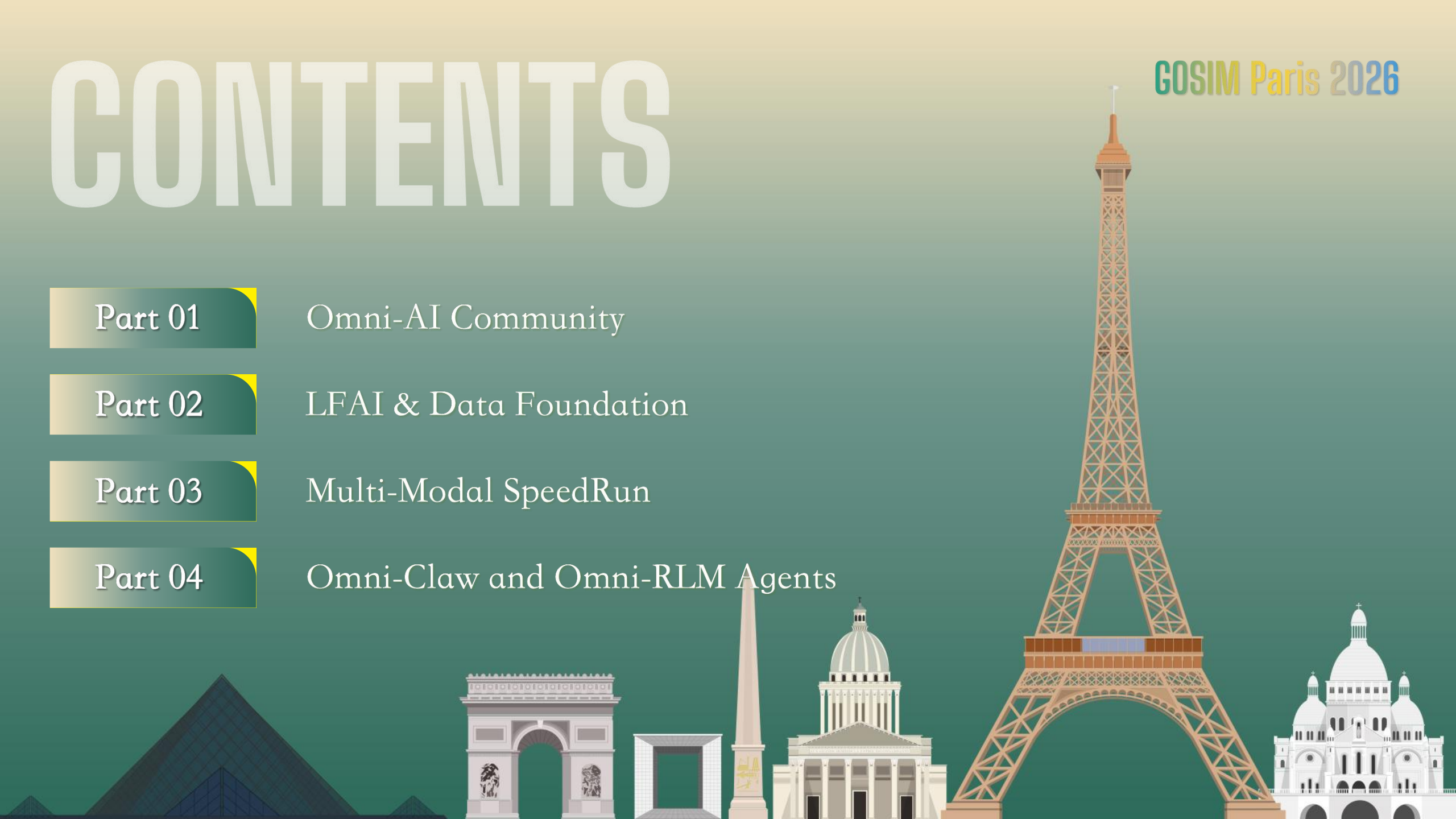

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01

Omni-AI Community

Omni-infer: An Enterprise-Grade LLM Inference Acceleration Toolkit for Ascend NPU

Inference Framework

Omni Proxy

Load Balance | Scale out

vLLM

- Scheduler
- Worker
- Page Attention
- Model Runner

NPU Platform

Omni NPU

Worker | model runner | attention | MOE

Inference Acceleration Suite

- API Server — Scale Out
- Omni EPLB — Layer wise | Uneven
- Omni Cache — APC | Offload
- MTP
- Model — Pangu | DeepSeek | Qwen
- Fusion Operator

Core Capabilities

Enterprise-Grade Low-Latency P/D Scheduling

Supports xPyD scheduling and horizontally scalable distributed Prefill/Decode (P/D) deployment, ensuring minimal latency at scale.

Request-Level Load Balancing

Optimizes both prefill and decode stages across varying sequence lengths to achieve maximum throughput with low latency.

Optimized MoE Expert Deployment

Supports large-scale Mixture-of-Experts (MoE) models with EP144/EP288 configurations.

MoE Expert Load Balancing

Enables hierarchical non-uniform redundancy and near real-time expert placement, improving overall resource utilization.

Advanced Attention Optimization

Tailored for LLMs, MLLMs, and MoE models to enhance performance and scalability.

Omni Cache

A DRAM-centric KV architecture with decoupled storage and computation, enabling high-capacity, high-performance, long-context inference.

Omni Proxy

Support for P/D Disaggregated Proxy Pipeline

Built on NGINX subrequest mechanisms, enabling Prefill-first then Decode execution, tailored for disaggregated inference architectures.

Stage-Aware Load Balancing

Prefill is optimized for TTFT, while Decode targets TPOT and throughput stability, enabling stage-specific optimization strategies.

Multi-Dimensional State-Aware Scheduling

Scheduling decisions are made by jointly considering queue status, KV cache pressure, and token-level workload.

DRAM-Centric KV Decoupled Architecture for Accelerated Inference (Omni Cache)

KV Storage-Compute Decoupling

Separates KV storage from computation based on workload characteristics, significantly reducing HBM memory requirements.

Continuous KV Buffer

Executes attention computation within contiguous memory buffers, improving operator efficiency.

Sparse Prefetching

Native sparse KV optimization reduces HBM usage by over 80% for long-context inference.

KV Fabric

Enables globally shared KV cache with orders-of-magnitude improvements in APC hit rate.

A Community Driven by Frontier Research and Open-Source Innovation

OMNIINFER: SYSTEM-WIDE ACCELERATION TECHNIQUES FOR OPTIMIZING LLM SERVING THROUGHPUT AND LATENCY

Jun Wang [1] Yunxiang Yao [1] Wenwei Kuang [1] Runze Mao [1] Zhenhao Sun [1] Zhuang Tao [1] Ziyang Zhang [1] Dengyu Li [1] Jiajun Chen [1] Zhili Wang [1] Kai Cui [1] Congzhi Cai [1] Longwen Lan [1] Ken Zhang* [1]

ABSTRACT

Large Language Models drive a wide range of modern AI applications but impose substantial challenges on large-scale serving systems due to intensive computation, strict latency constraints, and throughput bottlenecks. We introduce OmniInfer, a unified system-level acceleration framework designed to maximize end-to-end serving efficiency through fine-grained optimization of expert placement, cache compression, and scheduling. OmniInfer integrates three complementary components: OmniPlacement for load-aware Mixture-of-Experts scheduling, OmniAttn for sparse attention acceleration, and OmniProxy for disaggregation-aware request scheduling. Built atop vLLM, OmniInfer delivers system-wide performance gains through adaptive resource disaggregation, efficient sparsity exploitation, and global coordination across prefill and decode phases. Evaluated on DeepSeek-R1 within a 10-node Ascend 910C cluster, OmniInfer achieves 616 QPM, where the unified framework reduces TPOT by 36%, and the superimposition of OmniProxy further slashes TTFT by 38%. The project is open-sourced at https://gitee.com/omniai/omniinfer.

1 INTRODUCTION

Large Language Models (LLMs) like DeepSeek (Guo et al., 2025a), Qwen (Yang et al., 2025), Kimi (Team et al., 2025) and GPT (Agarwal et al., 2025) have rapidly become the foundation of modern AI applications, powering chatbots (Achiam et al., 2023), tool use (Schick et al., 2023), search (ZillizTech, 2025), and multimodal assistants (Sharon & Brichtova, 2025). However, their deployment at scale remains challenging due to enormous computational demands and stringent service-level objectives (SLOs), such as low time-to-first-token (TTFT) and high query-per-minute (QPM) throughput. To sustain practical serving, systems must not only maximize hardware utilization but also optimize scheduling across diverse and dynamic workloads.

Existing LLM serving systems have made substantial progress in improving inference efficiency, leveraging techniques such as operator fusion (Zuo et al., 2025), continuous batching (Kwon et al., 2023), and KV-cache reuse (Zheng et al., 2024). However, most of these systems assume homogeneous clusters and monolithic execution, where the prefill and decode stages are tightly coupled. This design overlooks the distinct computational and memory characteristics of the two phases: the prefill stage is compute-intensive, dominated by large matrix multiplications, whereas the decode stage is memory- and communication-bound, constrained by KV-cache accesses and sequential token generation (Patel et al., 2024; Zhong et al., 2024). Recent variants such as chunked prefill (Agrawal et al., 2024) attempt to reduce TTFT under continuous batching, but fundamentally shift rather than eliminate prefill–decode interference, and thus cannot fully resolve the inherent contention between the two stages. Treating the two phases uniformly not only results in inefficient hardware utilization but also exacerbates head-of-line (HOL) blocking under dynamic workloads (Zhou et al., 2024; Pan & Li, 2025).

To mitigate these inefficiencies, recent work has explored prefill–decode (PD) disaggregation (Zhong et al., 2024; Patel et al., 2024), which separates prefill and decode across different hardware resources. This approach enables finer-grained scheduling, better load balancing, and new opportunities for parallelization. Nonetheless, current PD disaggregation systems remain limited: many assume high-speed interconnects that are not universally available, while others lack global scheduling policies capable of adapting to fluctuating workloads. Consequently, while PD disaggregation represents an important step forward, it alone does not fully address the broader bottlenecks in modern LLM serving.

Beyond PD disaggregation, existing serving frameworks continue to face two fundamental challenges. First, *sparse computing*, arising from mixture-of-experts (MoE) models and sparse attention mechanisms, offers theoretical effi-

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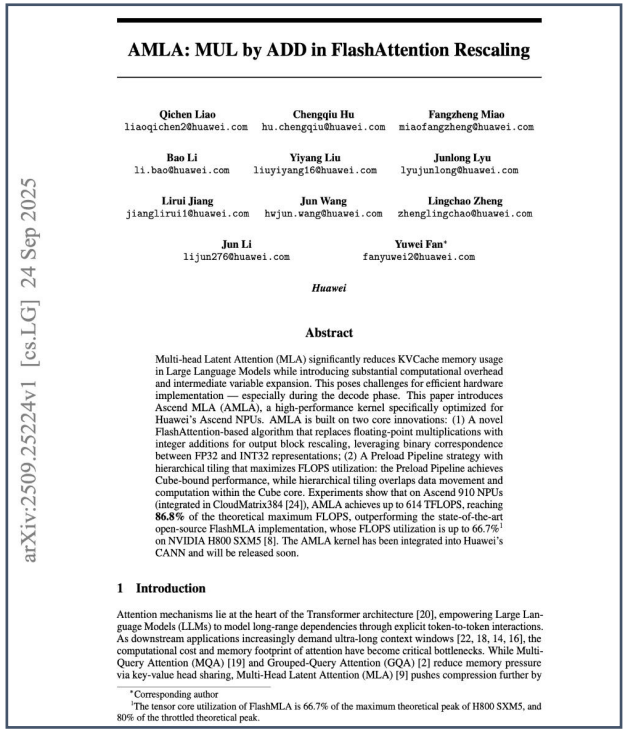
https://arxiv.org/abs/2511.22481



AMLA: MUL by ADD in FlashAttention Rescaling

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Abstract

Multi-head Latent Attention (MLA) significantly reduces KVCache memory usage in Large Language Models while introducing substantial computational overhead and intermediate variable expansion. This poses challenges for efficient hardware implementation — especially during the decode phase. This paper introduces Ascend MLA (AMLA), a high-performance kernel specifically optimized for Huawei's Ascend NPUs. AMLA is built on two core innovations: (1) A novel FlashAttention-based algorithm that replaces floating-point multiplications with integer additions for output block rescaling, leveraging binary correspondence between FP32 and INT32 representations; (2) A Preload Pipeline strategy with hierarchical tiling that maximizes FLOPS utilization: the Preload Pipeline achieves Cube-bound performance, while hierarchical tiling overlaps data movement and computation within the Cube core. Experiments show that on Ascend 910 NPUs (integrated in CloudMatrix384 [24]), AMLA achieves up to 614 TFLOPS, reaching **86.8%** of the theoretical maximum FLOPS, outperforming the state-of-the-art open-source FlashMLA implementation, whose FLOPS utilization is up to 66.7%[1] on NVIDIA H800 SXM5 [8]. The AMLA kernel has been integrated into Huawei's CANN and will be released soon.

1 Introduction

Attention mechanisms lie at the heart of the Transformer architecture [20], empowering Large Language Models (LLMs) to model long-range dependencies through explicit token-to-token interactions. As downstream applications increasingly demand ultra-long context windows [22, 18, 14, 16], the computational cost and memory footprint of attention have become critical bottlenecks. While Multi-Query Attention (MQA) [19] and Grouped-Query Attention (GQA) [2] reduce memory pressure via key-value head sharing, Multi-Head Latent Attention (MLA) [9] pushes compression further by

*Corresponding author

[1]The tensor core utilization of FlashMLA is 66.7% of the maximum theoretical peak of H800 SXM5, and 80% of the throttled theoretical peak.

https://arxiv.org/abs/2509.25224

Theoretically Optimal Attention/FFN Ratios in Disaggregated LLM Serving

Chendong Song [1] Meixuan Wang [2] Hang Zhou [3] Hong Liang [4] Yuan Lyu [4] Zixi Chen [5] Yuwei Fan [4] Zijie Zhou [1]

Abstract

Attention–FFN disaggregation (AFD) is an emerging architecture for LLM decoding that separates state-heavy, KV-cache–dominated Attention computation from stateless, compute-intensive FFN computation, connected by per-step communication. While AFD enables independent scaling of memory and compute resources, its performance is highly sensitive to the Attention/FFN provisioning ratio: mis-sizing induces step-level blocking and costly device idle time. We develop a tractable analytical framework for sizing AFD bundles in an rA–1F topology, where the key difficulty is that Attention-side work is nonstationary—token context grows and requests are continuously replenished with random lengths—while FFN work is stable given the aggregated batch. Using a probabilistic workload model, we derive closed-form rules for the optimal A/F ratio that maximize average throughput per instance across the system. A trace-calibrated AFD simulator validates the theory: across workloads, the theoretical optimal A/F ratio matches the simulation-optimal within 10%, and consistently reduces idle time.

1. Introduction

The explosive growth of large language models (LLMs) (Brown et al., 2020; Chowdhery et al., 2023; OpenAI, 2023; Kaplan et al., 2020) has created unprecedented challenges for efficient inference serving at scale. As model sizes expand into hundreds of billions of parameters, modern LLM serving inevitably requires distributed multi-device architectures. This reality has spurred a shift from monolithic serving paradigms toward increasingly sophisticated disaggregation strategies that decompose the inference pipeline across heterogeneous hardware resources.

Early efforts of disaggregate serving focused on stage-level separation, exemplified by the Prefill-Decode (PD) disaggregation (Zhong et al., 2024; Patel et al., 2024). LLM inference comprises two phases with distinct resource profiles: *prefill* processes the input prompt and generates the KV cache via dense matrix operations (compute-bound), while *decode* generates tokens autoregressively, reading the accumulated KV cache at each step (memory-bound). PD disaggregation separates these phases onto hardware matched to their respective bottlenecks. By disaggregating prefill and decode onto separate hardware pools that can be independently scaled, PD architectures enable higher decode batch sizes through request aggregation, improving GPU utilization during token generation.

Building upon this foundation, recent research has recognized that even within the decode phase, computational heterogeneity persists between different transformer components. The Attention-FFN Decoupling (AFD) (Wang et al., 2025; Zhang et al., 2025; Zhu et al., 2025; Zuo et al., 2025) extends this disaggregation further: Attention layers are stateful and memory-bound (dominated by KV cache reads), whereas FFN layers are stateless and can achieve compute-bound operation with sufficient batching. Traditional coupled architectures often leave FFN compute units underutilized due to the small decode batch with TPOT constraints. AFD architecture strategically disaggregates these components, allowing multiple Attention instances—which host the massive KV cache states—to feed into an aggregated FFN instances. This topology maximizes FFN arithmetic intensity while independently scaling memory resources for Attention.

Recent work (Wang et al., 2025; Zhu et al., 2025; Zuo et al., 2025) has shown that AFD can significantly improve serving efficiency by allowing Attention and FFN to scale independently. Given this flexibility, a central design decision in such systems is the ratio of Attention instances to FFN instances, which we denote as r. Currently, existing systems set r through empirical search, yet a theoretical foundation

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Preprint. January 30, 2026.

https://arxiv.org/abs/2601.21351

Uniting Global Developers to Build a Vibrant Community

Contributor

176

2026

Code	Open Source Community	Ecosystem Partners	Event	Sig
29万+	9	17	28	11

Community Activities

- Share & Gain Insights
- Solve Problems Together
- Learn from Practitioners

Offline Meetups have been successfully held in Shenzhen, Guangzhou, Beijing, Shanghai, and other cities across China

Omni-AI Website

Data as of April 30, 2026

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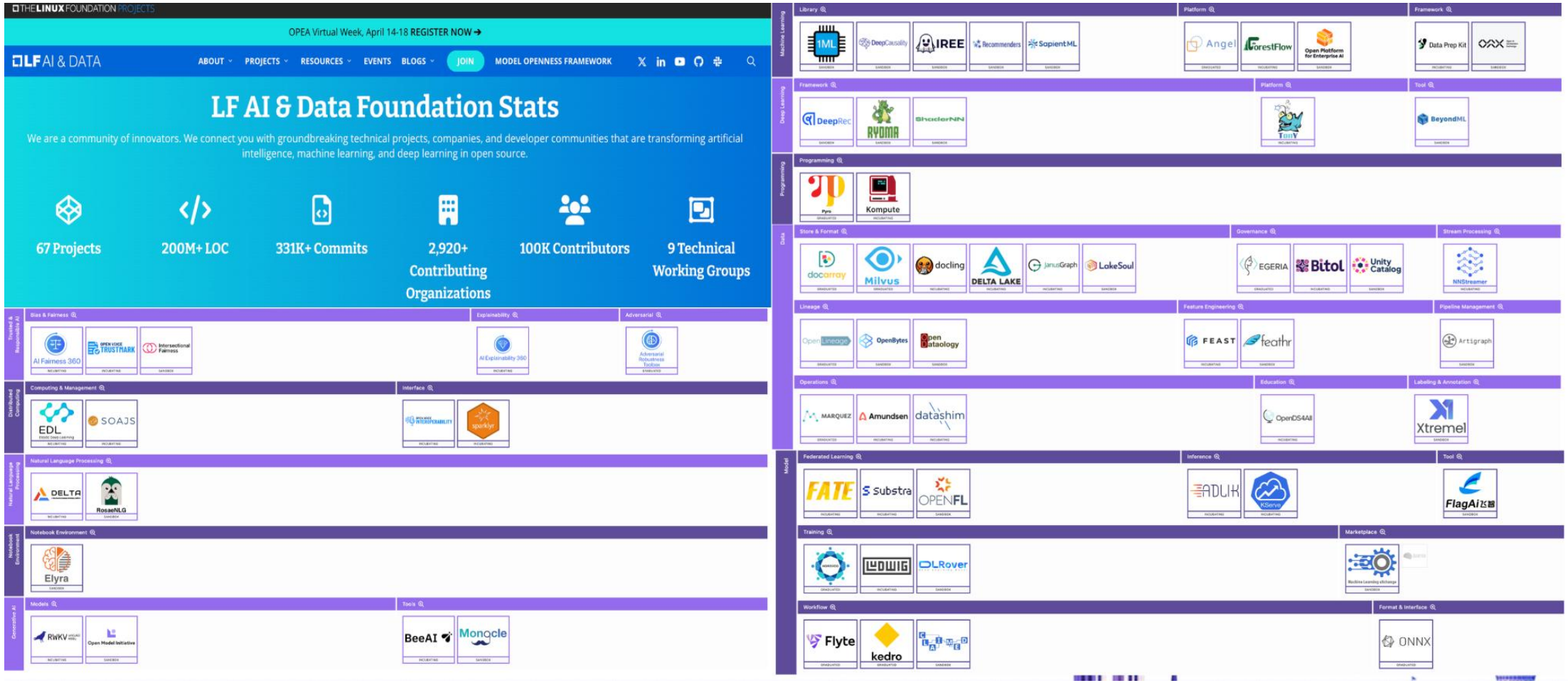
02

LFAI & Data Foundation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67 Projects
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331K+ Commits
2,920+ Contributing Organizations
100K Contributors
9 Technical Working Groups
Machine Learning
Deep Learning
Programming
Data
Model
Trusted & Responsible AI
Distributed Computing
Natural Language Processing
Notebook Environment
Generative AI
Library
Platform
Framework
Tool
Programming
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Governance
Stream Processing
Lineage
Feature Engineering
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Workflow
Format & Interface
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DeepRec
RYOMA
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BeyondML
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docarray
Milvus
docling
DELTA LAKE
JanusGraph
LakeSoul
EGERIA
Bitol
Unity Catalog
NNStreamer
OpenLineage
OpenBytes
Open Datology
FEAST
feathr
Artigraph
MARQUEZ
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datashim
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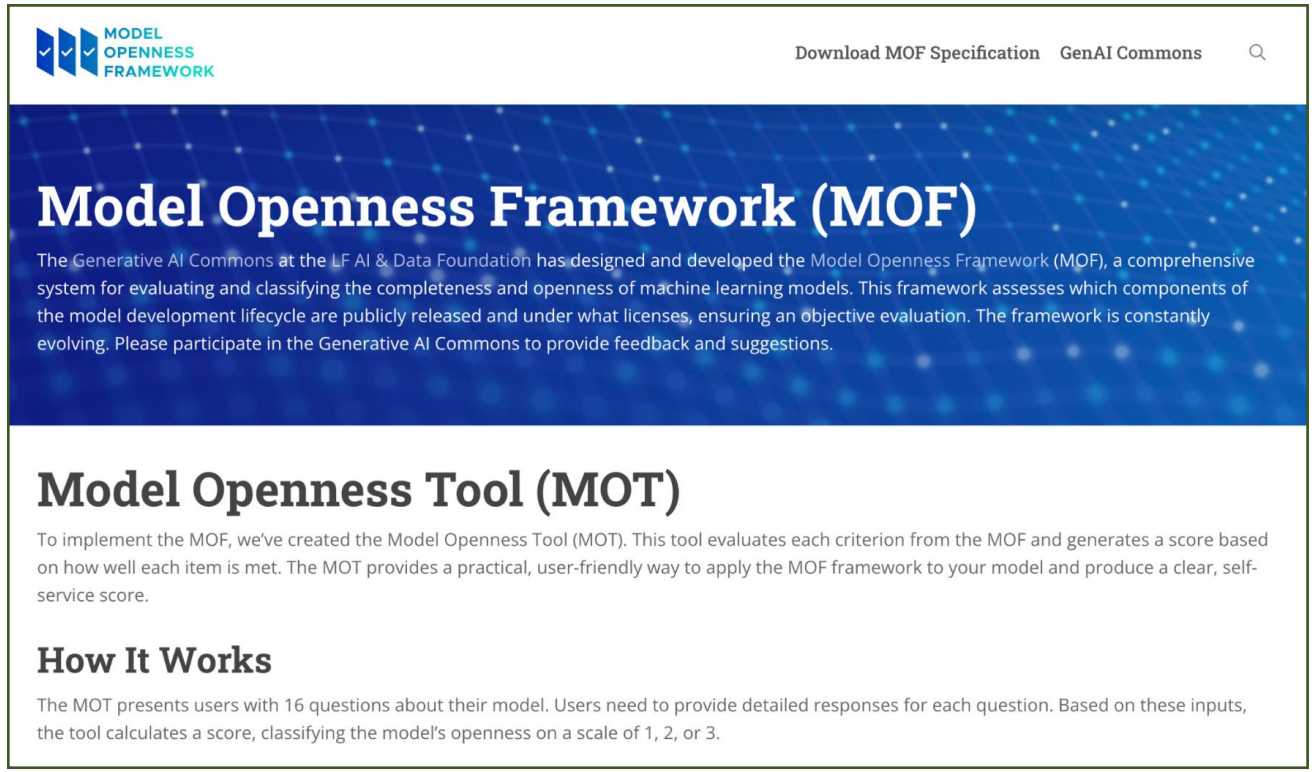
Substra
OPENFL
ADLIK
KServe
FlagAI
LEDWIG
DLRover
Flyte
kedro
ONNX
AI Fairness 360
TRUSTMARK
Intersectional Fairness
AI Explainability 360
Adversarial Robustness Toolbox
EDL
SOAJS
sparklyr
DELTA
RosaeNLG
Elyra
RWKV
Open Model Initiative
BeeAI
Monocle

Project	ONNX	Milvus	docling	DELTA LAKE	kedro
Website	ONNX.ai	Milvus.io	Docling.dev	Delta.io	Kedro.org
Focus	Machine Learning Models	Vector Database	Toolkit for Document Conversion	Storage Framework	Python Framework
Stats	• 44,037 Github stars • 8,357 contributors • 83,422 lifetime commits • 699 commits over last 30 days	• 36,932 Github stars • 1,271 contributors • 75,611 lifetime commits • 937 commits over last 30 days	• 36,503 Github stars • 1,891 contributors • 8,333 lifetime commits • 387 commits over last 30 days	• 13,206 Github stars • 1,056 contributors • 15,413 lifetime commits • 1,119 commits over last 30 days	• 11,842 Github stars • 1,724 contributors • 40,571 lifetime commits • 284 commits over last 30 days
Key Contributing Organizations	AMD, IBM, Intel, Microsoft, Nvidia	Anthropic, Databricks, Huawei, IBM, Zilliz	Amazon, IBM, Nirmata, Microsoft, Red Hat	Adobe, Databricks, Microsoft, Scribd, UC Berkeley	Amazon, Aneira Health, Axpo Group, QuantumBlack, Unit8 SA
Recent Accomplishments	Actively working on defining a common spec for Generative AI support v. 1.18.0 released in May, with features including: - Support for Python 3.13 - Wheels for Windows Arm64 - FLOAT4E2M1 and multi-device configuration support	Release of VDBBench 1.0, an open-source benchmark designed from the ground up to test vector databases under realistic production conditions. Release of Milvus 2.6, delivers breakthrough innovations across three critical areas: cost reduction, advanced search capabilities, and architectural improvements for massive scale.	Recent integration of Docling into Red Hat Enterprise Linux (RHEL) AI operating system The Research team behind IBM and Red Hat's InstructLab project used Docling to extract reams of information from targeted PDFs to train InstructLab's underlying AI models. A forthcoming extension of Docling's capabilities to handle more complex data types, including math equations, charts, and business forms	Release of Delta Lake 3.3.0, featuring DeltaSpark - a library that provides the necessary integration between Delta Lake and Apache Spark, allowing users to read, write, and manage Delta tables using Spark's APIs and capabilities. The introduction of Delta Kernel - a lightweight library that abstracts the complexities of interacting with Delta Lake, providing APIs for metadata handling, schema reading, and transaction log processing.	The Release of Kedro 1.0 - a stable, curated core that gives you confidence to build, and a modular, extensible ecosystem that empowers you to grow. Highlights include revamped DataCatalog, user experience improvements to namespaces, improvements to runners, and a polished, public API. Kedro-Viz will phase out its Experiment Tracking feature in the upcoming release of Kedro-Viz 11.0, with complete removal in version 12.0 due to low user adoption and the availability of robust alternatives like MLflow.
Commonalities	#ModelStandard #Models #Runtime #Hardware #AIAcceleration #ModelInference, **#ModelInteroperability**	#DBEngine #VectorDB #DataManagement #Scalability BatchAndStreaming**#VectorSimilaritySearch** #Data	**#AIDataPreprocessing** #Scalabilty #Data #AIDataPreparation	**#ReliableDataLakes** #DataGovernance, #Linage #Scalability #AIDataPreparation #Data	#Framework **#ReproducibleDataPipelines** #Data , #AIDataPreparation #BatchAndStreaming #Orchestration

12

*Organized consecutively by descending number of github stars as/of 8-19-25



Name	Organization	Classification	Last updated	Badge
Amber	LLM360	Class I - Open Science Model	2025-05-22	Class I - Open Science Model Qualified
Aquila-VL-2B	Beijing Academy of Artificial Intelligence (BAAI)	Class I - Open Science Model	2025-12-09	Class I - Open Science Model Qualified
OLMo-1B-hf	Allen Institute for AI	Class I - Open Science Model	2025-10-30	Class I - Open Science Model Qualified
OLMo-7B	Allen Institute for AI	Class I - Open Science Model	2025-11-12	Class I - Open Science Model Qualified
OLMo-7B-hf	Allen Institute for AI	Class I - Open Science Model	2025-10-30	Class I - Open Science Model Qualified
OLMo-7B-Twin-2T-hf	Allen Institute for AI	Class I - Open Science Model	2025-10-30	Class I - Open Science Model Qualified
Pythia-12B	EleutherAI	Class I - Open Science Model	2025-11-03	Class I - Open Science Model Qualified
Pythia-2.8B	EleutherAI	Class I - Open Science Model	2025-11-03	Class I - Open Science Model Qualified
Pythia-70M	EleutherAI	Class I - Open Science Model	2025-11-03	Class I - Open Science Model Qualified

LF Affiliated

Pre Training
(PyTorch Foundation)

Agent Protocol
(Agentic AI Foundation)

Cloud Native AI Infra
(CNCF Foundation)

Non-LF Affiliated

Post Training
(VeRL/Slime/JAX baked RL projects…)

Inference
(SGlang/TileRT/FastInfer/)

Kernel
(Tilelang/TK/Mirage/Apache TVM)

Agentic and RL
(Prime Intellect/Nous Research/Arcee AI/…)

Model and Optimization
(HuggingFace, marin, nanogpt-speedrun,...)

○ ○ ○

Agentic Infra
(Docling/Kedro/Milvus and more)

Open Model Initiative
(Omni-Modal Recipes/Benchmark and LoRA Standard)

Agentic Env
(to be constructed)

LFAI&Data Foundation

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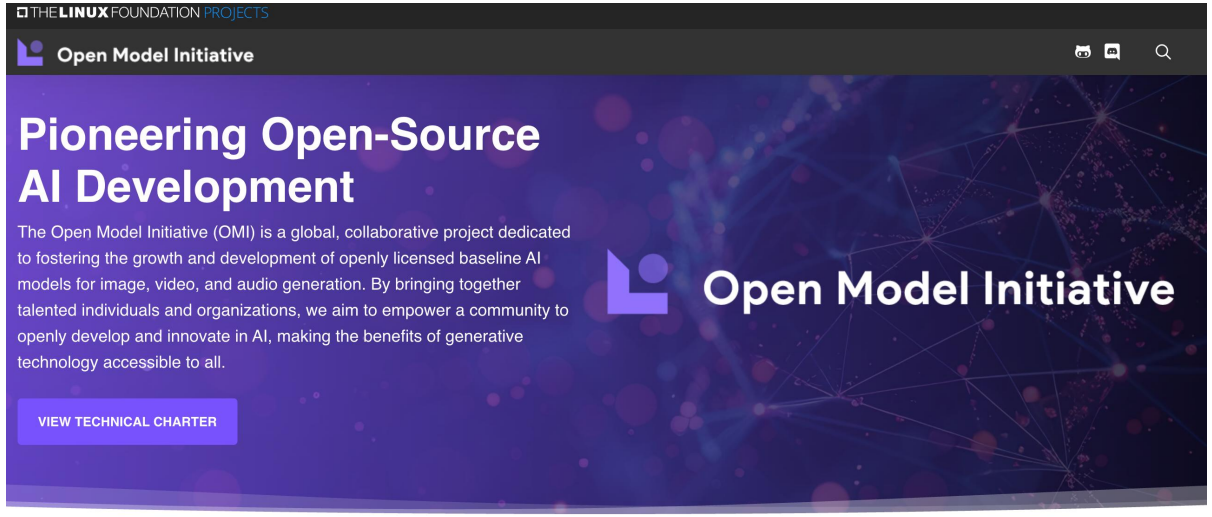
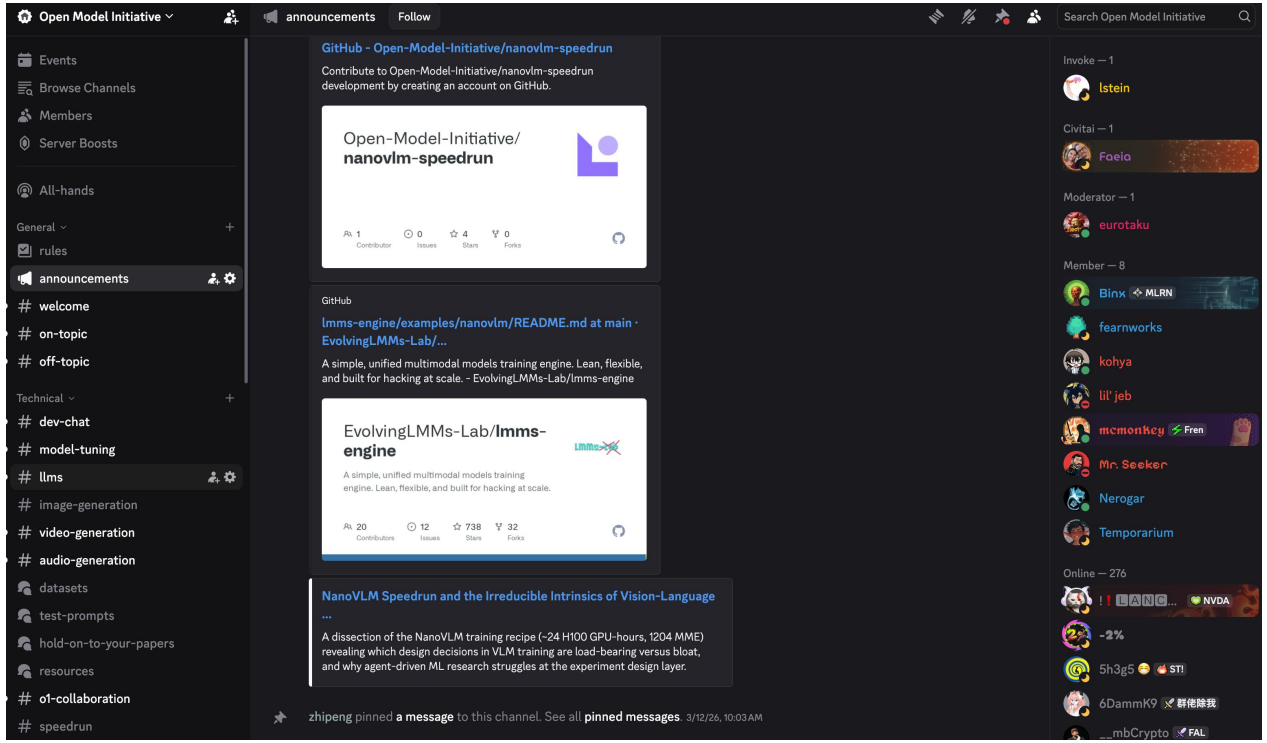
03

Multi-Modal SpeedRun

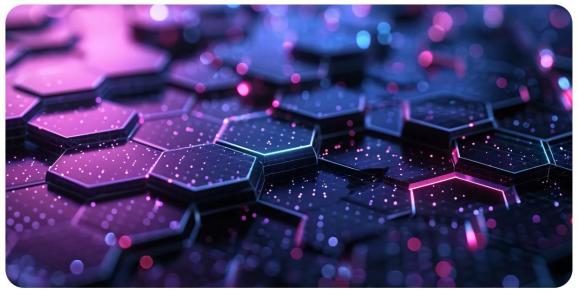




Our Vision and Mission

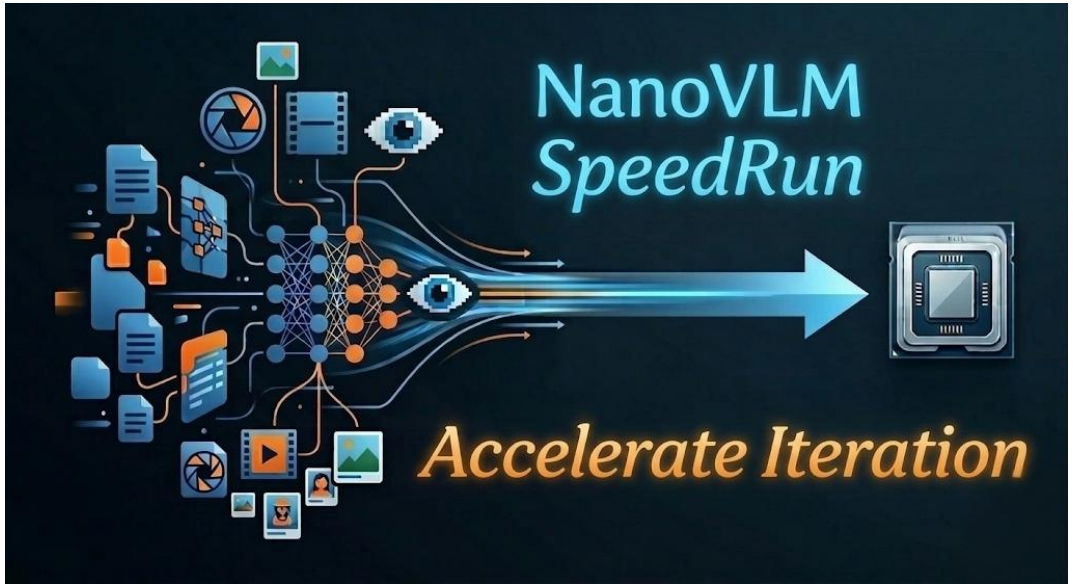
The Open Model Initiative is driven by a commitment to openness and collaboration. Our mission is to:

- Create and Maintain Openly Licensed AI Models
- Promote Responsible AI Development and Transparency
- Support Cross-Community and Cross-Project Collaboration



Define Problems, Set Rules, Build Baselines, and Collaborate on Benchmarking

https://github.com/Open-Model-Initiative/imagegen-speedrun/



https://github.com/Open-Model-Initiative/nanovlm-speedrun

imagegen-speedrun / speedrun-basic-rule-set.md

le Blame 108 lines (68 loc) · 2.31 KB

Primary Evaluation Curve

Each method must report:

```
Sample-wise Metric
^
|                       *
|                   *
|               *
|           *
|       *
|   *
+-----------------------------------> Wall Clock Time
```

- **x-axis**: Wall Clock Time
- **y-axis**: Sample-wise t2i metrics (GenEval1/2, HPSv3)

Note:

- The metric is open to discussion.

3. Dataset and Task

Item	Specification
Dataset	ImageNet (train split)
Resolution	256*256 resolution
Sampling steps (NFE)	100 (fixed across runs)

2821c4b SpeedRun-DiT-202601 / results.txt

SwayStar123 add xl results

Code Blame 460 lines (374 loc) · 15.1 KB

```
XL/1 @ 100k 2.99 it/s
Inception Score: 122.14485168457031
FID: 7.156954631595056
sFID: 4.976955193591721
Precision: 0.75316
Recall: 0.5972

XL/1 @ 400k 2.99 it/s
Inception Score: 178.141357421875
FID: 2.996033075112507
sFID: 4.411010295598999
Precision: 0.76946
Recall: 0.6317

    with SPRINT + RMS + ROPE + VALRES:
        Inception Score: 279.7873229980469
        FID: 3.101697263426729
        sFID: 4.9480099019050385
        Precision: 0.83302
        Recall: 0.5651

        kd_value: 0.24849
        kd_variance: 0.00434

        with RELU2:
            Inception Score: 265.0180969238281
            FID: 2.61406202232871
            sFID: 4.812557585819945
            Precision: 0.81848
            Recall: 0.5853
```

Primary Leaderboard

Each method must report a single leaderboard entry that includes both benchmark scores and training compute.

For the initial NanoVLM release, the leaderboard should contain:

Method	Train Time	Total FLOPs	MME	MMMU_Val	MMStar	GQA	ChartQA	DocVQ
NanoVLM Baseline	23.75 GPU hours (8 x H100)	335.02 PFLOPS	1204.46 (948.75/255.71)	0.3022	0.3273	0.4184	0.1084	0.1018

Reported benchmark metrics include:

- `MME` (Perception/Cognition)
- `MMBench` (EN Dev)
- `MMMU_Val`
- `MMStar` (Average)
- `GQA` (Exact Match)
- `ChartQA` (Relaxed Overall)
- `DocVQA` (ANLS)
- `OCRBench`
- `POPE` (Accuracy)

Item	Specification
Stage 1 data	`LMMs-Lab-Speedrun/Data_NanoVLM/Stage1-LLaVA-Pretrain`
Stage 2 data	`LMMs-Lab-Speedrun/Data_NanoVLM/Stage2-LLaVA-NeXT-Data`

The initial task is to train a compact VLM with the released two-stage recipe:

- **Stage 1**: projector-only alignment between vision and language
- **Stage 2**: instruction tuning on the released multimodal data recipe

Note:

- Local YAML path configuration is allowed, but the underlying dataset content and split policy should remain unchanged.
- Any major dataset change should be proposed as a separate track rather than silently changing the baseline.

Baseline Recipe

The initial NanoVLM SpeedRun baseline uses:

- **LLM**: `Qwen/Qwen3-0.6B`
- **Vision Encoder**: `google/siglip2-so400m-patch16-naflex`
- **Projector**: LLaVA-style 2-layer MLP
- **Training Paradigm**: 2-stage training

Model Changes

- The baseline recipe is the default comparison point
- Participants may propose architecture or training changes
- Any deviation from the baseline recipe must be clearly reported
- Total trainable parameter count must be reported

Note:

- Future versions may split submissions into a strict baseline track and a more open innovation track.

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Stage	Frozen	LR	Batch	Steps	PFLOPS	H100-hrs
1 (Alignment)	vision + LLM	10^{-3}	32×8	2180	236.79	19.32
2 (Instruction)	vision only	2×10^{-5}	8×8	11540	98.23	4.43

Total: ~335 PFLOPS, ~24 H100-hours. MME: 1204.46. Data: 558K caption pairs (Stage 1), 738K filtered instruction samples (Stage 2).

Training Intrinsics

Intrinsic 1: The Vision Encoder Stays Frozen

SigLIP2-so400m stays frozen throughout both stages. This follows the standard LLaVA two-stage recipe: freeze the vision encoder, train only the projector (Stage 1), then unfreeze the LLM while keeping the vision encoder frozen (Stage 2). The practical motivation is straightforward - the vision encoder is already a strong feature extractor out of the box, and fine-tuning it on 558K captioning pairs risks degrading features learned from much larger pretraining data, while also increasing GPU memory requirements significantly.

Whether this is strictly necessary is an open question. Some VLM recipes (e.g., PaLI, InternVL) do fine-tune the vision encoder at larger data scales and report gains. At the NanoVLM compute budget, freezing is the conservative and validated choice.

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The Agent Research Gap

Karpathy recently observed that AI agents running nanochat optimization experiments produce poor experiment designs. They vary parameters without controlling for compute, generate spurious results (e.g., "discovering" that larger hidden dimensions lower validation loss - trivially true in the infinite-data regime), and fail to create strong baselines before ablating.

NanoVLM illustrates exactly what makes this hard. The recipe above contains dozens of coupled decisions - which components to freeze, learning rate ratios, batch sizes per stage, data filtering thresholds. Each decision interacts with every other. An agent that independently varies hidden dimension will produce results confounded by uncontrolled variables.

The deeper problem: training intrinsics are not in the loss landscape. You cannot gradient-descend your way to "freeze the vision encoder." You need meta-knowledge that vision encoders are already well-trained and that 558K samples cannot improve them. This is the kind of tacit knowledge that Karpathy's nanochat miniseries makes explicit through iso-FLOP methodology - always compare at fixed compute, use task-level metrics (CORE) not raw validation loss.

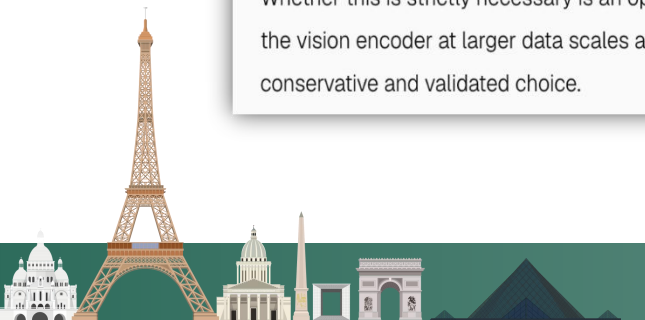
Formalizing the Design Space

Can we make training intrinsics navigable? Consider a decision DAG:

```
vision_encoder -> freeze_strategy -> projector_design -> stage1_lr
              \-> data_budget -> stage1_data (alignment)
                             \-> stage2_data (instruction, filtered)
              \-> compute_budget -> batch_per_stage -> memory_constraint
```

Each node has a small set of valid configurations. Edges encode constraints (if vision encoder is frozen, projector must be trainable). An agent navigating this DAG with iso-FLOP baselines at each node could systematically discover NanoVLM-quality recipes.

The question is whether this DAG can be learned from the literature, or whether it requires the kind of judgment that accumulates from years of failed training runs.



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04

Omni-Claw and Omni-RLM Agents

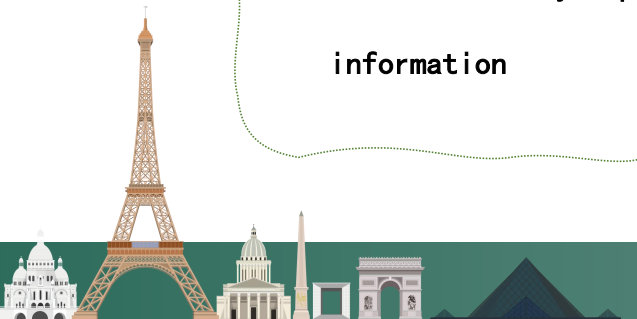
Why Long-Text Reasoning Matters

"Traditional LLMs try to process everything in one pass, but complex tasks often require going back and forth through the information. This motivates approaches like recursive reasoning systems."

- AI works well for answering questions and summarizing short content
- But real-world information is often **long, complex, and spread across many sources**
- Solving these tasks requires **breaking problems into smaller steps and revisiting context**
- This points to a new approach: systems that can **iteratively explore and reason over information**

Here comes with RLM

- "Breaking problems into smaller steps" → recursive decomposition
- "Revisiting context" → iterative / multi-pass reasoning
- "Explore and reason" → hints at agent-like + REPL-style interaction



A High-Performance Recursive AI Agent Framework

Core Capability

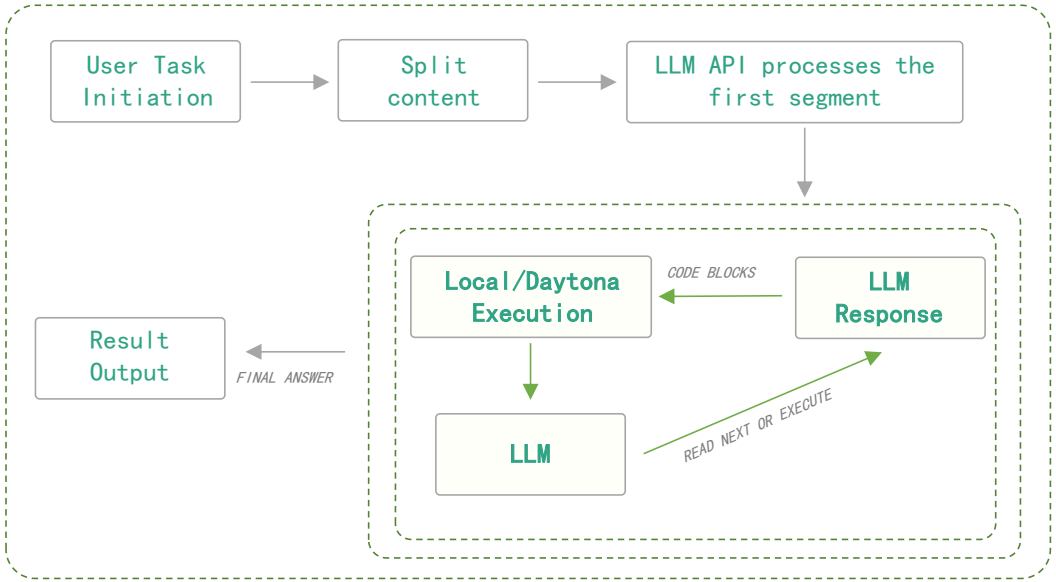
- **Recursive Infer Engine:**
Performs iterative recursive reasoning to progressively approach the final answer.
- **Long-Context Segmentation:**
Splits long text into segments and processes them recursively via sandboxed content variables.
- **Dual Execution Environments:**
Supports both local Python execution and Daytona sandbox environments.
- **End-to-End Observability:**
Structured logging enables full traceability and reproducibility of the execution process.

Rewrite by Zig

- Type Safety
- Memory Safety
- High Performance
- Easy Packaging

Powerful RLM

- Decompose Complex Problems
- Solve Step by Step



Recursive, Stop until final answer found

Use Cases

- Complex Problems
- Tool-Using Agents
- Automated Decomposition

Lightweight AI Agent Runtime Built with Zig

LLM-Powered Planning

Accepts natural language input, leverages Omni-RLM for intent understanding and tool selection, and automatically generates structured execution plans.

Extensible Tool Registry

Provides built-in `exec` and `finish` tools with support for custom extensions. Tools are registered as functions and invoked dynamically by the model.

Multi-LLM API Compatibility

Follows the OpenAI Chat Completions standard, enabling seamless integration with OpenAI, Moonshot, local Ollama, and other compatible services.

Interactive REPL Consol

Provides a command-line conversational interface with real-time input and output display, including built-in commands such as `/config` and `/tools` for management.

Minimalist High-Perf Runtime

Built with Zig, featuring zero GC, low memory footprint, single-binary deployment, and no additional runtime dependencies

Result

```
> 用git看看这个项目最近4次更新都更新了什么
$ git log -4 --oneline

=== Task Completed ===
Final answer: 这个项目最近4次更新的内容如下： 1. 更新了Omni_RLM依赖到0.1.2版本，并重构了运行时的对话日志处理。 2. 增强初始化以支持
Agent和Runtime的自定义最大迭代次数。 3. 可能是针对拉取请求查找的修复。 4. 添加了自动删除对话日志的功能。
Total tool calls: 1
  1. exec -> success (git log -4 --oneline)
> 我的电脑系统是什么，还有多少电量，能用多久
$ uname -a && upower -i /org/freedesktop/UPower/devices/battery_BAT0
$ sw_vers && pmset -g batt

=== Task Completed ===
Final answer: 您的电脑系统是macOS，版本为26.3.1。当前电池电量为99%，剩余使用时间为19小时09分钟。
Total tool calls: 2
  1. exec -> failed (uname -a && upower -i /org/freedesktop/UPower/devices/battery_BAT0)
  2. exec -> success (sw_vers && pmset -g batt)
```

Technical Report & Code Bundle
https://gitcode.com/ascend-tribe/ascend-inference-cluster

LFAI&Data Foundation/Open Model Initiative
https://github.com/Open-Model-Initiative/

Source Code Development & Collaboration
https://gitee.com/omniai/omniinfer

NanovIm SpeedRun
https://github.com/Open-Model-Initiative/nanovlm-speedrun

Community Governance & Operations
https://gitee.com/omniai/community

SpeedRun AI-CoScientist
https://coscientist.lmms-lab.com/p/cs/qLTGKSaKjCcnoB7zI_bcU

Github
https://github.com/omni-ai-npu/omni-infer

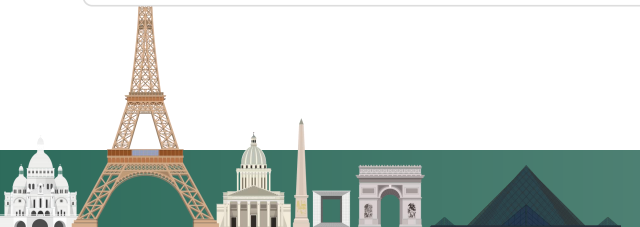
ImageGen SpeedRun
https://github.com/Open-Model-Initiative/imagegen-speedrun

Openl Community
https://git.openi.org.cn/omni-ai/omni-infer

Omni-Claw
https://github.com/Open-Model-Initiative/omni-claw

GitLink Platform
https://gitlink.org.cn/omni-ai/omniinfer

Omni-RLM
https://github.com/Open-Model-Initiative/Omni-RLM



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Thanks!